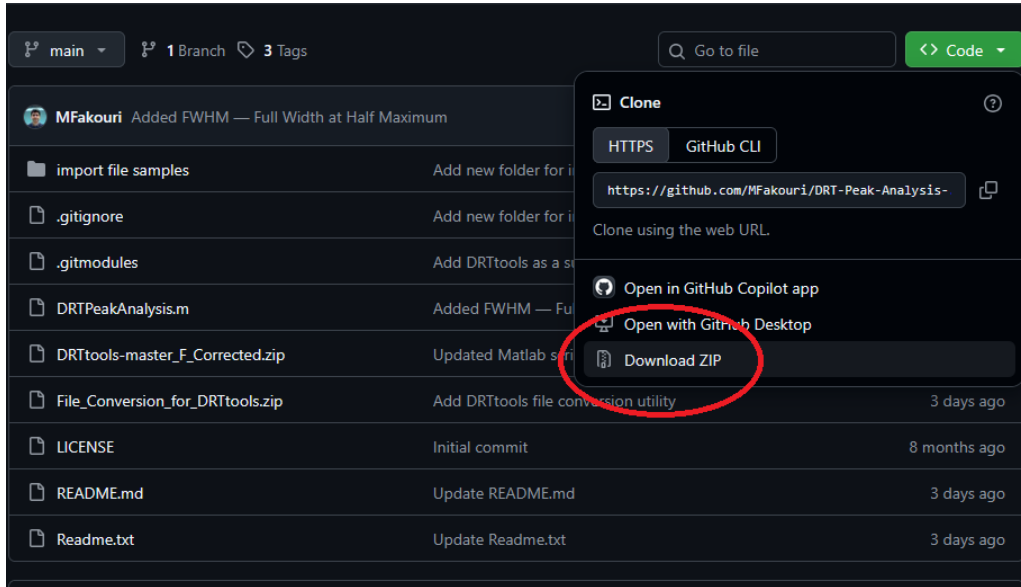


DRT-Peak-Analysis User Guide

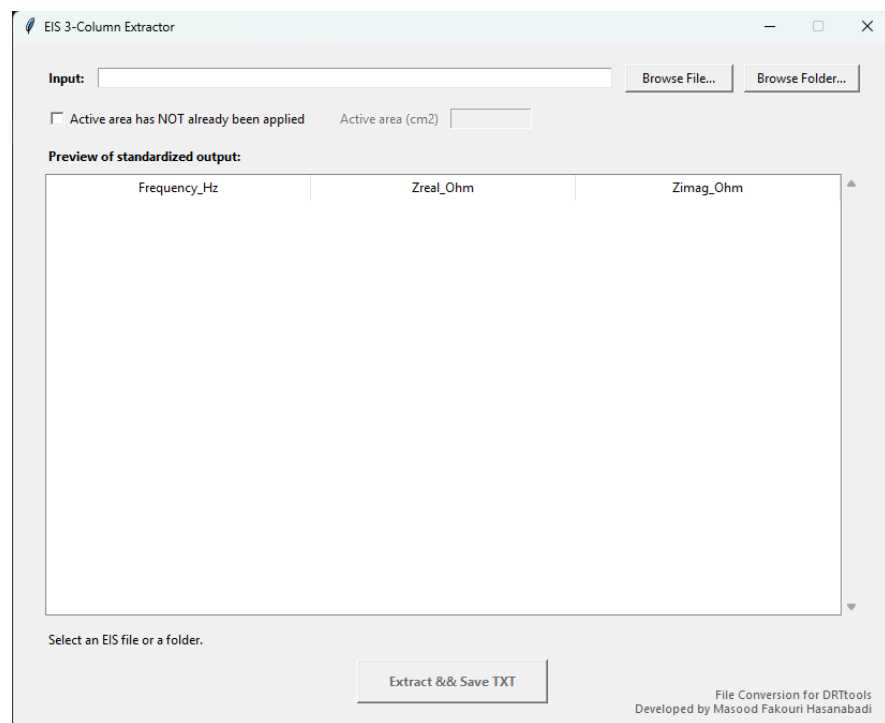
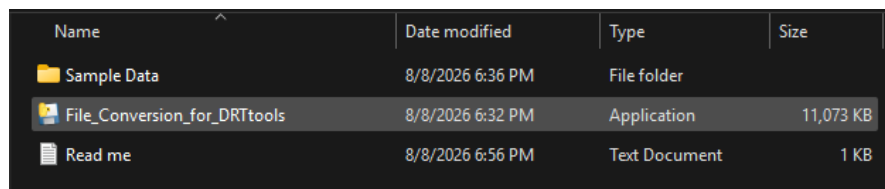
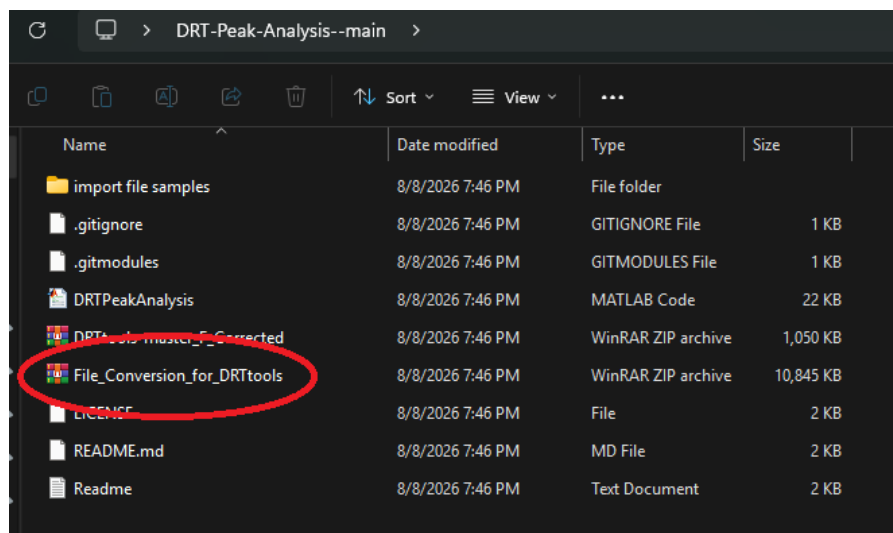
1. Download the latest version of the code from the following link:

<https://github.com/MFakouri/DRT-Peak-Analysis-.git>

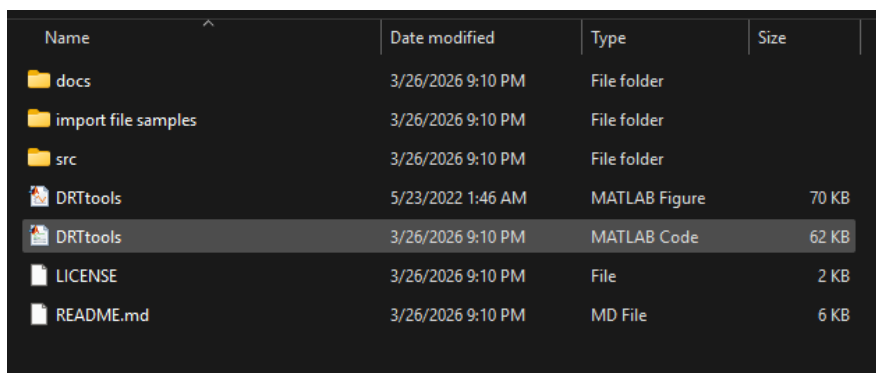
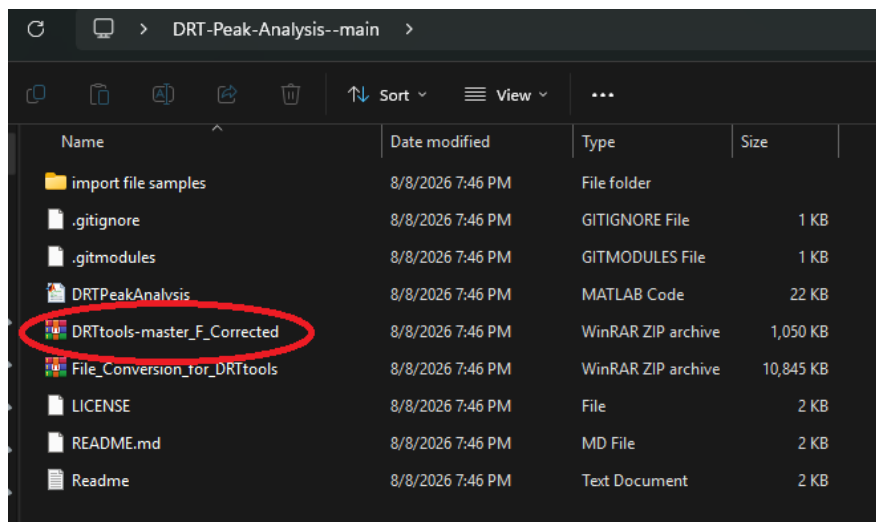


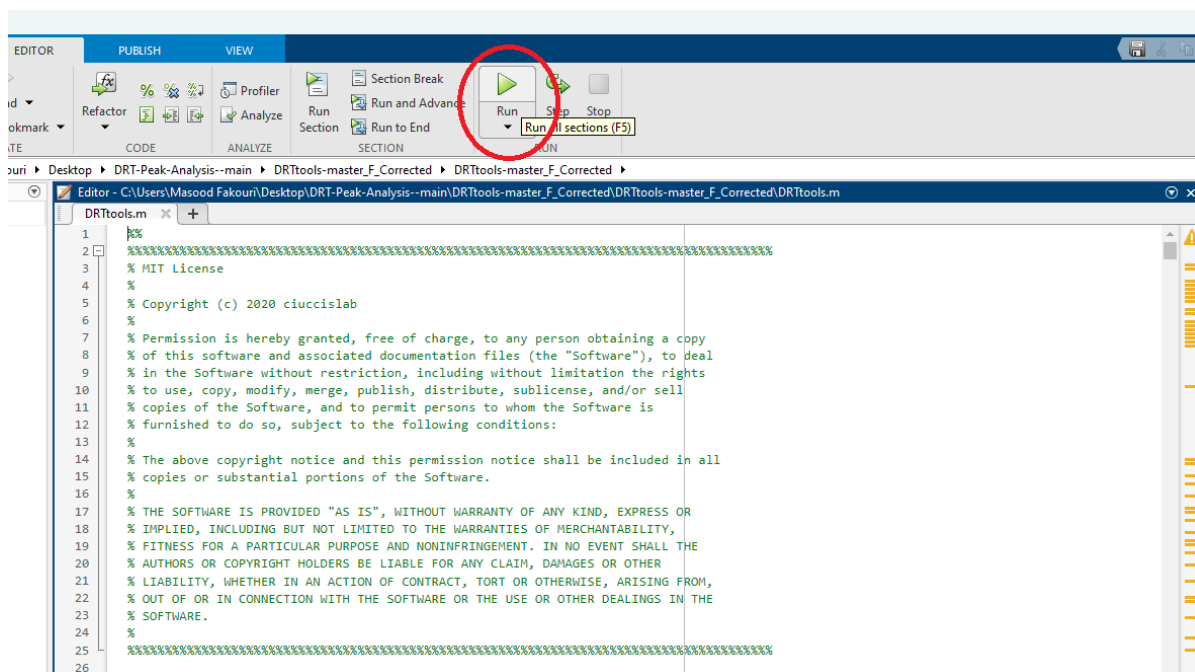
2. Unzip the folder.

3. Unzip the File Conversion for DRTtools utility to convert the original output files into the format required by DRTtools. Data from different instruments, including Solartron, Autolab, Gamry, and Zahner, can be converted. However, the data must first be saved in TXT format using the respective instrument software. If you have a folder containing several EIS data files, select "Select Folder" to convert all files in a single run.



4. A modified version of DRTtools is available in the “DRTtools-master_F_Corrected” folder. Unzip the folder to start the DRT analysis. Then, run the DRTtools MATLAB code.





4. Import one of the converted files.

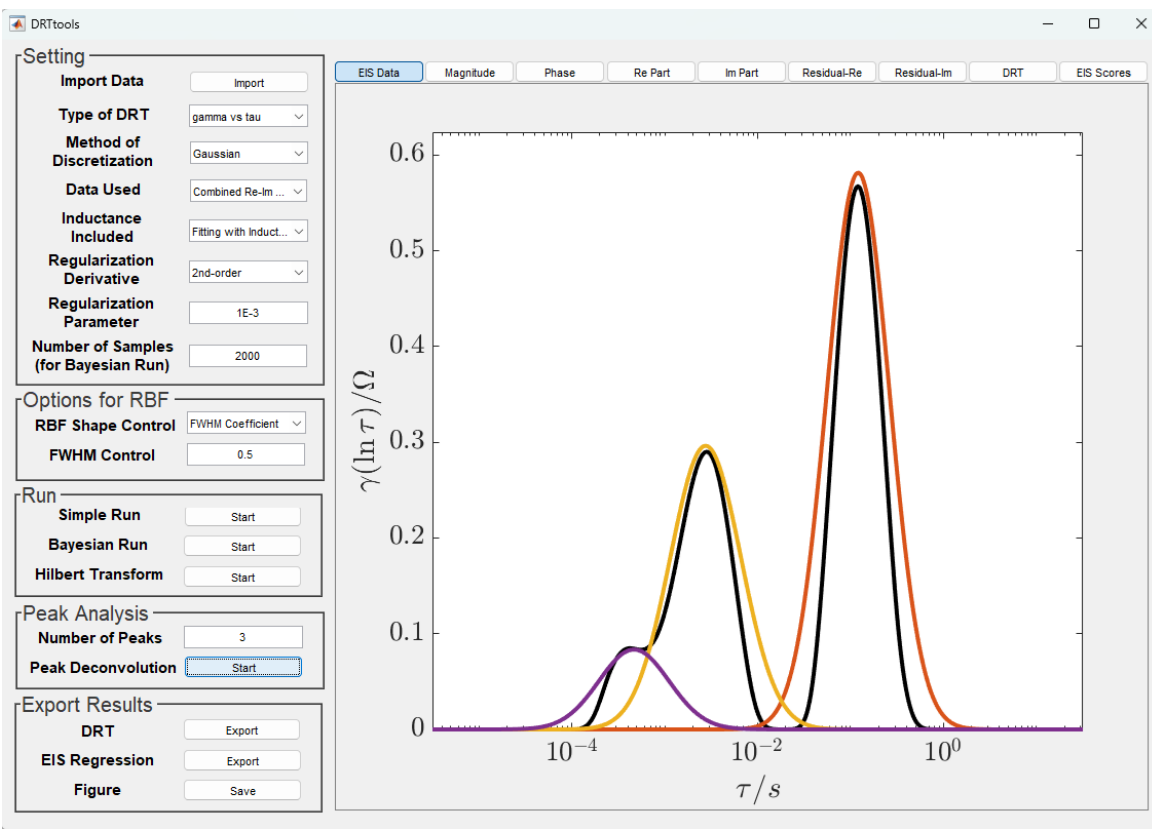
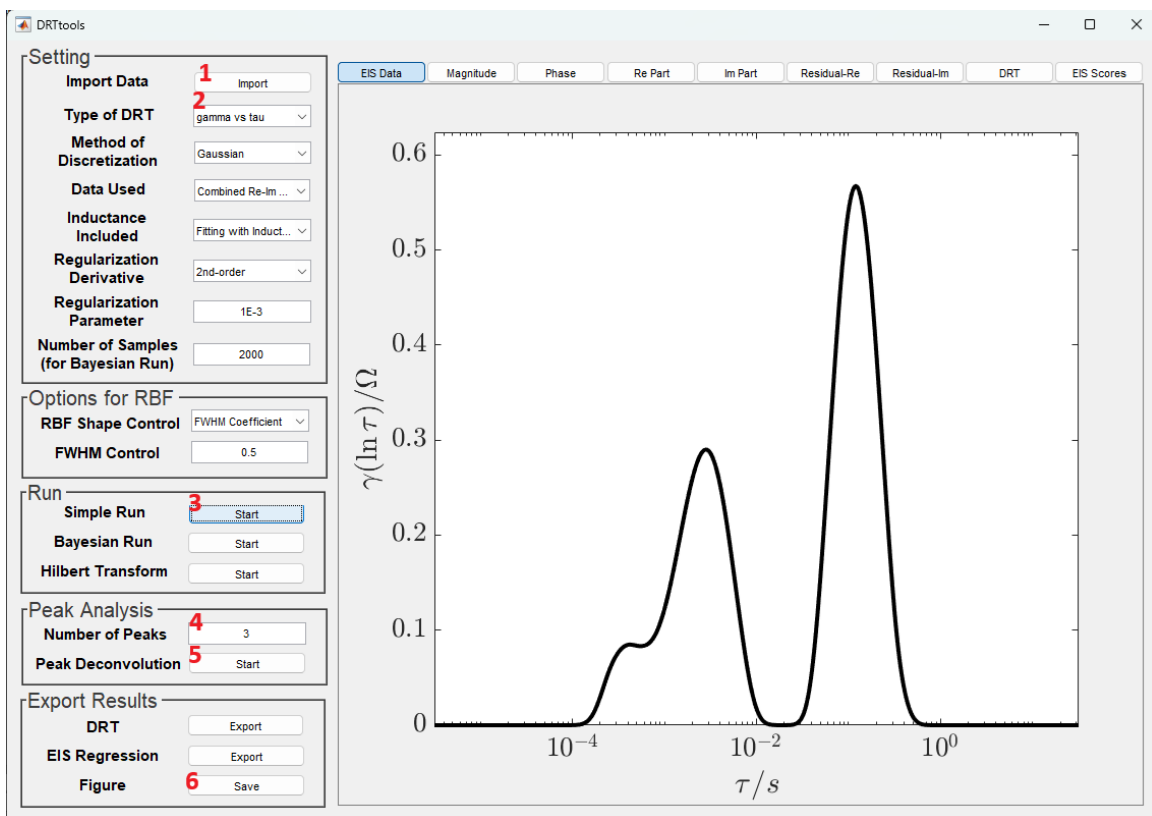
Keep "Type of DRT" set to "Gamma vs Tau." You may adjust the other parameters to modify the results.

Click "Simple Run" to start the DRT analysis.

Select the number of peaks visible in the DRT plot.

Click "Start Peak Deconvolution" to perform peak deconvolution.

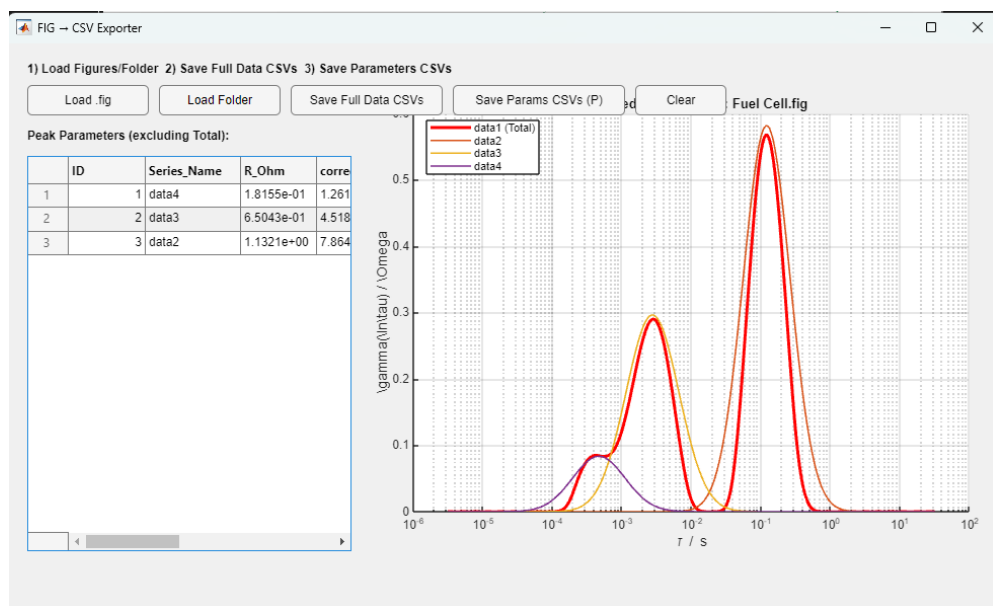
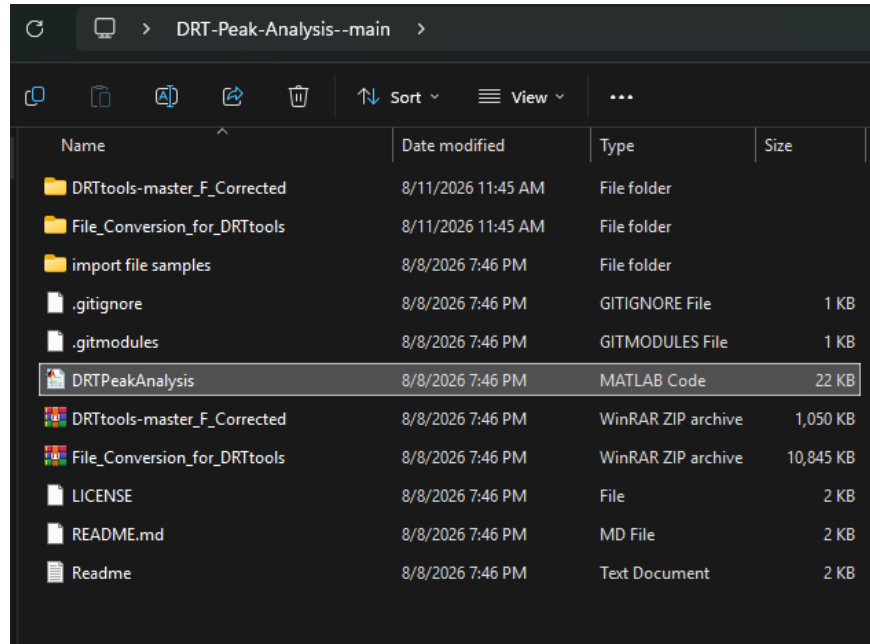
Click "Save Figure" to save the resulting figure.



5. Run DRTPeakAnalysis.

Load a file or a folder containing multiple files.

You can save the analysis parameters or the individual data for each peak.



	A	B	C	D	E	F	G	H
	data1_X	data1_Y	data2_X	data2_Y	data3_X	data3_Y	data4_X	data4_Y
	3.16E-06	2.29E-52	3.16E-06	4.96E-41	3.16E-06	3.16E-14	3.16E-06	5.48E-09
	3.26E-06	4.46E-52	3.26E-06	8.53E-41	3.26E-06	4.16E-14	3.26E-06	6.73E-09
	3.36E-06	8.54E-52	3.36E-06	1.47E-40	3.36E-06	5.46E-14	3.36E-06	8.25E-09
	3.47E-06	1.60E-51	3.47E-06	2.51E-40	3.47E-06	7.16E-14	3.47E-06	1.01E-08
	3.58E-06	2.96E-51	3.58E-06	4.30E-40	3.58E-06	9.38E-14	3.58E-06	1.24E-08
	3.69E-06	5.37E-51	3.69E-06	7.35E-40	3.69E-06	1.23E-13	3.69E-06	1.51E-08
	3.81E-06	9.55E-51	3.81E-06	1.25E-39	3.81E-06	1.60E-13	3.81E-06	1.84E-08
	3.93E-06	1.67E-50	3.93E-06	2.14E-39	3.93E-06	2.09E-13	3.93E-06	2.24E-08
	4.05E-06	2.87E-50	4.05E-06	3.63E-39	4.05E-06	2.72E-13	4.05E-06	2.73E-08
	4.18E-06	4.84E-50	4.18E-06	6.17E-39	4.18E-06	3.54E-13	4.18E-06	3.32E-08
	4.31E-06	8.01E-50	4.31E-06	1.05E-38	4.31E-06	4.61E-13	4.31E-06	4.02E-08
	4.45E-06	1.30E-49	4.45E-06	1.77E-38	4.45E-06	5.98E-13	4.45E-06	4.88E-08
	4.59E-06	2.08E-49	4.59E-06	2.99E-38	4.59E-06	7.75E-13	4.59E-06	5.90E-08
	4.74E-06	3.27E-49	4.74E-06	5.05E-38	4.74E-06	1.00E-12	4.74E-06	7.14E-08
	4.88E-06	5.04E-49	4.88E-06	8.51E-38	4.88E-06	1.30E-12	4.88E-06	8.62E-08
	5.04E-06	7.64E-49	5.04E-06	1.43E-37	5.04E-06	1.68E-12	5.04E-06	1.04E-07
	5.20E-06	1.14E-48	5.20E-06	2.40E-37	5.20E-06	2.16E-12	5.20E-06	1.25E-07
	5.36E-06	1.67E-48	5.36E-06	4.03E-37	5.36E-06	2.79E-12	5.36E-06	1.50E-07
	5.53E-06	2.39E-48	5.53E-06	6.75E-37	5.53E-06	3.58E-12	5.53E-06	1.81E-07
	5.71E-06	3.29E-48	5.71E-06	1.12E-36	5.71E-06	4.61E-12	5.71E-06	2.17E-07

X-Y data of each peak.

A	B	C	D	E	F	G	H
ID	Series_Name	R_Ohm	corrected_R_Ohm	Tau_s	Capacitance_F	Frequency_Hz	FWHM_InTau
1	data4	0.1815535	0.126121476	0.00046934	0.002585109	339.1068083	2.044432614
2	data3	0.6504343	0.451843244	0.00275594	0.004237076	57.74978639	2.062411072
3	data2	1.1321071	0.786451309	0.12182651	0.107610417	1.306406464	1.828960987
4	Total Sum	1.9640949	1.36441603	NaN	NaN	NaN	NaN

Polarization resistance (R), capacitance, FWHM (Full Width at Half Maximum), and peak frequency/relaxation time (τ) for each peak of the DRT curve.

Contact and Citation

For questions, comments, or technical issues regarding this software, please contact:

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If you use this software in your research, please cite the GitHub repository using the “Cite this repository” option available on GitHub:

GitHub: <https://github.com/MFakouri/DRT-Peak-Analysis-.git>

Please cite the software in any publications or presentations where it is used.